

- * Forgetfulness, confusion
- * Hyperexcitability, nervousness
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth
- Rebound insomnia when withdrawing from long-term treatment



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait

- To avoid problems with memory, take estazolam only if planning to have a full night's sleep
- Lower the dose
- Switch to a shorter-acting sedative hypnotic
- Switch to a non-benzodiazepine hypnotic
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 1–2 mg/day at bedtime

Dosage Forms

- Tablet 1 mg scored, 2 mg scored

How to Dose

- Initial 1 mg/day at bedtime; increase to 2 mg/day at bedtime if ineffective



Dosing Tips

- Use lowest possible effective dose and assess need for continued treatment regularly
- Estazolam should generally not be prescribed in quantities greater than a 1-month supply
- Patients with lower body weights may require lower doses
- Risk of dependence may increase with dose and duration of treatment

Overdose

- No death reported in monotherapy; sedation, slurred speech, poor coordination, confusion, coma, respiratory depression

Long-Term Use

- Not generally intended for long-term use
- Evidence of efficacy up to 12 weeks

Habit Forming

- Estazolam is a Schedule IV drug
- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

How to Stop

- If taken for more than a few weeks, taper to reduce chances of withdrawal effects
- Patients with seizure history may seize upon sudden withdrawal
- Rebound insomnia may occur the first 1–2 nights after stopping
- For patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization

Pharmacokinetics

- Half-life 10–24 hours
- Inactive metabolites



Drug Interactions

- Increased clearance and thus decreased estazolam levels in smokers
- Increased depressive effects when taken with other CNS depressants (see Warnings below)



Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used

together, including specifically the risk of slowed or difficulty breathing and death

- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy
- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur
- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea
- Estazolam should be administered only at bedtime

Do Not Use

- If patient is pregnant
- If patient has angle-closure glaucoma

- If there is a proven allergy to estazolam or any benzodiazepine

Special populations

Renal Impairment

- Drug should be used with caution

Hepatic Impairment

- Drug should be used with caution

Cardiac Impairment

- Benzodiazepines have been used to treat insomnia associated with acute myocardial infarction

Elderly

- No dose adjustment in healthy patients
- Debilitated patients: recommended initial dose of 0.5 mg/day



Children and Adolescents

- Safety and efficacy have not been established
- Long-term effects of estazolam in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored



Pregnancy

- Contraindicated for use in pregnancy
- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy

Breast Feeding

- Unknown if estazolam is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Effects on infant have been observed and include feeding difficulties, sedation, and weight loss

The art of psychopharmacology

Potential Advantages

- Transient insomnia

Potential Disadvantages

- Smokers (may need higher dose)

Primary Target Symptoms

- Time to sleep onset
- Total sleep time
- Nighttime awakenings

Pearls

- If tolerance develops, it may result in increased anxiety during the day and/or increased wakefulness during the latter part of the night
- Best short-term use is for less than 10 consecutive days, and for less than half of the nights in a month
- Drug holidays may restore drug effectiveness if tolerance develops
- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment

Suggested Reading

Pierce MW , Shu VS . Efficacy of estazolam. The United States clinical experience . *Am J Med* 1990 ;**88** :S6 –11.

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Pierce MW , Shu VS , Groves LJ . Safety of estazolam. The United States clinical experience . *Am J Med* 1990 ;**88** :S12 –17.

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Vogel GW , Morris D . The effects of estazolam on sleep, performance, and memory: a long-term sleep laboratory study of elderly insomniacs . *J Clin Pharmacol* 1992 ;**32** :647–51 .

Eszopiclone

Therapeutics

Brands

- Lunesta
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Non-benzodiazepine hypnotic; alpha 1 isoform selective agonist of GABA-A/benzodiazepine receptors

Commonly Prescribed for

(bold for FDA approved)

- **Insomnia**
- Primary insomnia
- Chronic insomnia
- Transient insomnia

- Insomnia secondary to psychiatric or medical conditions
- Residual insomnia following treatment with antidepressants

How the Drug Works

- May bind selectively to a subtype of the benzodiazepine receptor, the alpha 1 isoform
- May enhance GABA inhibitory actions that provide sedative hypnotic effects more selectively than other actions of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as

obstructive sleep apnea or restless leg syndrome, which requires independent evaluation

- Increase the dose
- Improve sleep hygiene
- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- * Chronic studies of eszopiclone suggest lack of notable tolerance or dependence developing over time

Notable Side Effects

* Unpleasant taste

- Sedation
- Dizziness
- Dose-dependent amnesia
- Nervousness
- Dry mouth, headache



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with other CNS depressants in overdose
- Rare angioedema

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Next-day carryover sedation following nighttime dosing uncommon

What to Do About Side Effects

- Wait
- To avoid problems with memory, take eszopiclone only if planning to have a full night's sleep
- Lower the dose
- Switch to a shorter-acting sedative hypnotic
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 2–3 mg at bedtime

Dosage Forms

- Tablet 1 mg, 2 mg, 3 mg

How to Dose

- No titration, take dose at bedtime



Dosing Tips

- Not restricted to short-term use

- No notable development of tolerance or dependence seen in studies up to 6 months
- Recent study adding eszopiclone to patients with major depression and only a partial response to fluoxetine showed improvement not only in residual insomnia, but in other residual symptoms of depression as well
- Most studies were done with 3 mg dose or less at night, but some patients with insomnia associated with psychiatric disorders may require higher dosing
- However, doses higher than 3 mg may be associated with carryover effects, hallucinations, or other CNS adverse effects
- To avoid problems with memory or carryover sedation, only take eszopiclone if planning to have a full night's sleep
- Most notable side effect may be unpleasant taste
- Other side effects can include sedation, dizziness, dose-dependent amnesia, nervousness, dry mouth, and headache

Overdose

- Few reports of eszopiclone overdose, but probably similar to zopiclone overdose
- Rare fatalities have been reported in zopiclone overdose
- Symptoms associated with zopiclone overdose include clumsiness, mood changes, sedation, weakness, breathing trouble, unconsciousness

Long-Term Use

- No development of tolerance was seen in studies up to 6 months

Habit Forming

- Eszopiclone is a Schedule IV drug
- Some patients could develop dependence and/or tolerance with drugs of this class; risk may be theoretically greater with higher doses
- History of drug addiction may theoretically increase risk of dependence

How to Stop

- Rebound insomnia may occur the first night after stopping
- If taken for more than a few weeks, taper to reduce chances of withdrawal effects

Pharmacokinetics

- Metabolized by CYP450 3A4 and 2E1
- Terminal elimination half-life approximately 6 hours
- Heavy high-fat meal slows absorption, which could reduce effect on sleep latency



Drug Interactions

- Increased depressive effects when taken with other CNS depressants
- Inhibitors of CYP450 3A4, such as nefazodone and fluvoxamine, could increase plasma levels of eszopiclone
- Inducers of CYP450 3A4, such as rifampicin, could decrease plasma levels of eszopiclone

Other Warnings/Precautions

- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with caution in patients with impaired respiratory function or obstructive sleep apnea
- Eszopiclone should only be administered at bedtime
- Rare angioedema has occurred with sedative hypnotic use and could potentially cause fatal airway obstruction if it involves the throat, glottis, or larynx; thus if angioedema occurs treatment should be discontinued
- Sleep driving and other complex behaviors, such as eating and preparing food and making phone calls, have been reported in patients taking sedative hypnotics; in some cases these behaviors

have resulted in serious injury or death, prompting the FDA to require a black box warning

Do Not Use

- If the patient has experienced an episode of complex sleep behaviors after taking a sleep medication
- If there is a proven allergy to eszopiclone or zopiclone

Special populations

Renal Impairment

- Dose adjustment not generally necessary

Hepatic Impairment

- Dose adjustment not generally recommended for mild-to-moderate hepatic impairment
- For severe impairment, recommended initial dose 1 mg at bedtime; maximum dose 2 mg at bedtime

Cardiac Impairment

- Dosage adjustment may not be necessary

Elderly

- May be more susceptible to adverse effects

- Initial dose 1 mg at bedtime; maximum dose generally 2 mg at bedtime

Children and Adolescents

- Safety and efficacy have not been established
- Long-term effects of eszopiclone in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Infants whose mothers took sedative hypnotics during pregnancy may experience some withdrawal symptoms
- Neonatal flaccidity has been reported in infants whose mothers took sedative hypnotics during pregnancy

Breast Feeding

- Unknown if eszopiclone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

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Potential Advantages

- Primary insomnia
- Chronic insomnia
- Those who require long-term treatment
- Those with depression whose insomnia does not resolve with antidepressant treatment

Potential Disadvantages

- More expensive than some other sedative hypnotics

Primary Target Symptoms

- Time to sleep onset
- Nighttime awakenings
- Total sleep time

* May be preferred over benzodiazepines because of its rapid onset of action, short duration of effect, and safety profile

- Eszopiclone is the best documented agent to be safe for long-term use, with little or no suggestion of tolerance, dependence, or abuse
- May even be safe to consider in patients with a past history of substance abuse who require treatment with a hypnotic
- May be preferred over benzodiazepine hypnotics, which all cause tolerance, dependence, and abuse as a class
- Not a benzodiazepine itself but binds to the benzodiazepine receptor
- May be a preferred agent in primary insomnia
- Targeting insomnia may prevent the onset of depression and maintain remission after recovery from depression
- Rebound insomnia does not appear to be common

Suggested Reading

Eszopiclone: esopiclone, estorra, S-zopiclone, zopiclone–Sepracor . *Drugs R D* 2005 ;**6** (2):111–15 .

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Krystal AD , Walsh JK , Laska E , et al. Sustained efficacy of eszopiclone over 6 months of nightly treatment: results of a randomized, double-blind, placebo-controlled study in adults with chronic insomnia . *Sleep* 2003 ;**26** (7):793 –9 .

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Zammit GK , McNabb LJ , Caron J , Amato DA , Roth T . Efficacy and safety of eszopiclone across 6-weeks of treatment for primary insomnia . *Curr Med Res Opin* 2004 ;**20** (12):1979 –91.

Flibanserin

Therapeutics

Brands

- Addyi

see index for additional brand names

Generic?

No

Class

- Serotonin 1A agonist and serotonin 2A antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Acquired, generalized hypoactive sexual desire disorder (HSDD) in premenopausal women**

How the Drug Works

- Sexual dysfunction is theoretically linked to an imbalance in central excitatory and inhibitory sexual signals
- HSDD hypothetically results from excessive inhibitory signals, inadequate excitatory signals, or a combination of the two

- Flibanserin hypothetically counteracts this imbalance in HSDD through its ability to both reduce inhibitory signals and enhance excitatory signals
- Specifically, flibanserin increases the release of dopamine and norepinephrine, which are excitatory sexual signals, and reduces the release of serotonin, an inhibitory sexual signal

How Long Until It Works

- In clinical trials, improvement was seen at 4 weeks

If It Works

- Increases number of satisfying sexual events and ratings on sexual desire scores
- Reduces distress related to sexual dysfunction

If It Doesn't Work

- If there is no improvement after 8 weeks, discontinue use



Best Augmenting Combos for Partial Response or Treatment Resistance

- None known
- Theoretically, bupropion augmentation could be considered, but no published trials

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Most common side effects (e.g., notable side effects below) likely caused by 5HT1A agonism and/or 5HT2A antagonism
- Hypotension, dizziness, and syncope of flibanserin as a monotherapy is theoretically related to its 5HT1A agonist actions
- Weak alpha 1 antagonist actions may also contribute to hypotension and dizziness of flibanserin, especially when given in combination with alcohol, which is contraindicated

Notable Side Effects

- Somnolence
- Nausea
- Fatigue
- Insomnia
- Dry mouth



Life-Threatening or Dangerous Side Effects